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# [Short Paper] Forensic Analysis of Indirect Prompt Injection Attacks on LLM Agents

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***Abstract*—Large language model (LLM) agents are vulnerable to a range of evolving attacks including Indirect Prompt Injection (IPI). Digital investigations involving IPI attacks on LLM agents are challenging due to growing data volumes and agent flow complexity. We introduce a novel approach for forensic analysis of LLM agent execution logs to identify IPI attacks. We implement a custom digital forensic analysis framework featuring a realistic agentic task benchmark to generate agent logs, for subsequent analysis using 12 state-of-the-art LLMs. Our preliminary results show promising potential for malicious trail detection, with varying performance across domain-specific test suites. We discuss key findings, limitations, and future work directions to support the empirical evaluation of LLMs for digital forensic applications.**

***Index Terms*—LLM, agent, indirect prompt injection, digital forensics**

## I. Introduction

Large language model (LLM) agents provide a set of capabilities to perform complex tasks, reason about problems and generate and execute solution plans, via tool interactions with the broader environment [1]. Despite extensive research into the associated detections and defences, LLM-integrated applications, including those featuring LLM agents, remain vulnerable to a wide range of evolving attacks [2]. LLM inputs can combine data and instructions from a variety of sources, including external systems. Indirect inputs handled by LLM agents enable dangerous Indirect Prompt Injection (IPI) attacks, because they can allow adversaries to exploit LLM agent functionality without having a direct interface to an LLM-integrated application. To mount IPI attacks, adversaries plant crafted malicious prompts that deceive LLMs into generating outputs designed to bypass intended agent functionality, when processed by the associated model [3].

In light of growing criminal case backlogs and rapidly increasing data volume and associated complexity to foster forensic investigations [4], IPI attacks targeting LLM agents present additional challenges for digital forensic (DF) investigators [5]. DF practitioners will require empirically validated digital evidence identification and analysis methods to streamline such investigations. Prior research suggests that LLMs show strong potential to aid with some of these challenges and enhance the associated DF activities [6].

This paper proposes a novel approach for automated identification and analysis of potential DF evidence contained in LLM agent execution logs. The key contributions of this work include the following:

- Introduce the DF analysis approach based on LLM agent execution logs as a potential DF evidentiary source, and
- Quantify the utility of selected state-of-the art LLMs for assisting with DF analysis of LLM agent execution logs.

## II. Related Work

Usage of LLMs to support and enhance DF investigations is an active research area. LLMs have shown significant potential for various DF activities such as forensic artefact identification, analysis, code generation and DF task automation [4]–[6], [8]–[10].

Researchers explored the utility of `GPT-4` for DF search keyword and code generation, whilst noting limitations around lack of determinism and support for large data volumes and binary payloads [6]. To enhance the DF process model with AI, ChatGPT Forensics Analysis – Evidence Identification (CFA-EI) was proposed as a novel human-in-the loop AI-driven framework based on `ChatGPT` [8]. However, the authors of CFA-EI emphasised the need for empirical validation and human verification of generated outputs. Similarly, LLMs such as `LLama-2` and `GPT-3.5` showed potential for DF report generation, but would also require output validation by DF experts [9]. Specific DF branches such as memory forensics could benefit from using LLMs for triage automation. An LLM-based approach for ransomware analysis in memory dumps achieved an average detection rate of 94%, showing significant potential for DF task automation [10]. LLMs also showed promising results for automating log analysis and anomaly detection [11], [12]. Incorporation of LLM agents to support DF activities as part of an orchestrated workflow also holds potential for streamlining DF investigations [5]. However, challenges associated with output non-determinism, potential biases, and explainability would need to be addressed to support wider acceptance by DF practitioners.

Despite significant promise, more research is needed to quantify and explore the utility of LLMs for specific DF activities, tasks and investigative contexts.

## III. Experiments and Forensic Analysis

Our proposed forensic analysis approach is depicted in Fig. 1. We focus on using a set of state-of-the-art LLMs to analyse agent prompts and responses generated by executing

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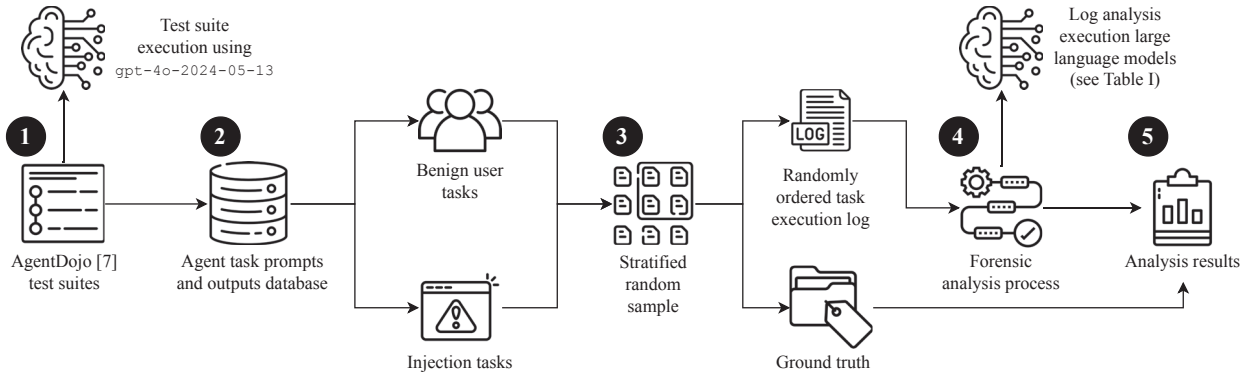



Fig. 1. Digital forensic analysis of AgentDojo [7] LLM agent task execution logs collected by capturing LLM invocation prompts and associated outputs.

LLM agent tasks for potential evidence of malicious prompts, which may contain prompt injection attacks.

To validate our approach, we implement a custom experimental testbed based on `AgentDojo` [7] as the basis for generating the data set. `AgentDojo` provides a comprehensive set of 97 realistic tasks across various business contexts coupled with 629 security test cases used to evaluate LLM agent prompt injection attacks and defences. Specifically, we use `gpt-4o-2024-05-13` and `important_instructions_no_names` non-adaptive IPI attack to execute the benign and injection tasks, as this model and attack type are expected to provide the highest model utility and attack success rate. We then adopt the following experimental flow:

1) Execute the `AgentDojo` benchmark sequentially across the four built-in test suites ("Slack", "Banking", "Travel" and "Workspace") covering both user and injection tasks.
2) Our fork of `AgentDojo` is augmented to collect LLM agent execution logs by capturing all LLM invocations prompts (inputs) and responses (outputs) to local persistent store. Model invocations triggered by agents also contain custom metadata including agent task run identifiers, test suite names and corresponding injections for injection tasks used to produce ground truth.
3) For each test suite, we apply stratified random sampling to select a subset of LLM agent execution logs, including all of the benign user tasks as well as logs from randomly selected injection tasks making up to 20% of benign tasks. We then randomly order LLM agent task run logs to avoid concentration of benign or injection tasks in close proximity and capture the output in a structured file based on JavaScript Object Notation (JSON). We also separately capture the ground truth used for the subsequent identification of log entries pertinent to injection tasks.
4) We analyse the resulting log files following an approach similar to the active analysis method described in [11], using a tailored analysis prompt targeting a selection of LLMs listed in Table I. During the analysis, we use the `temperature` value of `0.0` and iterate over the LLM agent log by filling the LLM context window up to the maximum supported size until the entire log file is analysed by each model being evaluated.
5) Finally, we analyse the outputs from the previous step against the ground truth to assess the utility of each LLM for this particular DF analysis task. We treat log lines that are associated with an injection task and correctly identified by an LLM as true positives, whereas identified log lines not associated with an injection task are treated as false positives.

Following the completion of our sequential analysis process for each of the four `AgentDojo` test suites, we also repeat this approach using a random sample selected across all four suites and a larger resulting data set called "Combined". Our preliminary results are shown in Table I.

## IV. Discussion

### A. Findings

LLM utility for agent trace analysis task is represented using precision (P), recall (R) and F1 (F1) scores for each test suite and model. In the context of our experiment, LLM utility varies greatly depending on the test suite and model. For instance, `claude-3-opus` and `gpt-4o` demonstrate consistent performance across multiple test suites. Other models such as `gemini-1.5-flash` and `llama3⋆` models show significant variability across different suites. Suites such as "Travel" and "Banking" did not yield any analysis findings by LLMs such as `claude-3-haiku`, `mistral-large-2`, `gpt-4o-mini` and `gpt-4-turbo`, whereas `gemini-1.5-flash` did not produce interpretable structured outputs for the "Travel" suite. The latter is likely because model responses were limited by inherent safety guardrails beyond our control. The impact of model context window size (W), which varies greatly between 8K and 1M for different LLMs, is also inconsistent. For example, `gemini-1.5-flash` shows outstanding performance on selected suites, with much lower scores results on others.



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TABLE I
DIGITAL FORENSIC ANALYSIS OF LLM AGENT LOGS USING SELECTED LLMS. T = MODEL TEMPERATURE, W = MAXIMUM LOG CONTEXT WINDOW SIZE TO FILL, P = PRECISION, R = RECALL, F1 = F1 SCORE. N = LLM INVOCATION COUNT, I = INJECTION COUNT.

| Model / Test suite | T | W | Slack (N=123,I=29) | | | Banking (N=65,I=29) | | | Travel (N=117,I=18) | | | Workspace (N=147,I=29) | | | Combined (N=444,I=88) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 |
| claude-3-opus-20240229 | 0.0 | 200K | 0.67 | 0.40 | 0.50 | 0.80 | 1.00 | 0.89 | 1.00 | 0.50 | 0.67 | 0.71 | 0.62 | 0.67 | **1.00** | **0.60** | **0.75** |
| claude-3-sonnet-20240229 | 0.0 | 200K | 0.75 | 0.60 | 0.67 | 0.00 | 0.00 | 0.00 | 1.00 | 0.25 | 0.40 | 0.33 | 0.25 | 0.29 | 0.67 | 0.30 | 0.41 |
| claude-3-5-sonnet-20240620 | 0.0 | 200K | 0.67 | 0.80 | 0.73 | 0.80 | 1.00 | 0.89 | 0.60 | 0.75 | 0.67 | 0.44 | 0.50 | 0.47 | 0.69 | 0.45 | 0.55 |
| claude-3-haiku-20240307 | 0.0 | 200K | 0.50 | 0.40 | 0.44 | 0.67 | 0.50 | 0.57 | 0.00 | 0.00 | 0.00 | 0.40 | 0.25 | 0.31 | 0.30 | 0.30 | 0.30 |
| mistral-large-2 | 0.0 | 128K | 0.06 | 0.60 | 0.11 | 0.10 | 1.00 | 0.18 | 0.00 | 0.00 | 0.00 | 0.07 | 0.25 | 0.11 | 0.12 | 0.10 | 0.11 |
| llama3-70b | 0.0 | 8K | 0.11 | 1.00 | 0.20 | 0.18 | 1.00 | 0.31 | 0.67 | 1.00 | 0.80 | 0.22 | 1.00 | 0.36 | 0.14 | 0.90 | 0.24 |
| llama3.1-405b | 0.0 | 64K | 0.43 | 0.60 | 0.50 | 0.40 | 0.50 | 0.44 | 1.00 | 0.75 | 0.86 | 0.33 | 0.75 | 0.46 | 0.46 | 0.55 | 0.50 |
| llama3.1-70b | 0.0 | 128K | 0.80 | 0.80 | 0.80 | 0.80 | 1.00 | 0.89 | 1.00 | 0.25 | 0.40 | 0.11 | 0.12 | 0.12 | 0.09 | 0.60 | 0.15 |
| gemini-1.5-flash | 0.0 | 1M | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | - | - | - | 0.11 | 0.75 | 0.19 | 0.21 | 0.50 | 0.30 |
| gpt-4o-2024-08-06 | 0.0 | 128K | 1.00 | 0.60 | 0.75 | 0.38 | 0.75 | 0.50 | **1.00** | **1.00** | **1.00** | **1.00** | **0.62** | **0.77** | 0.56 | 0.45 | 0.50 |
| gpt-4o-mini-2024-07-18 | 0.0 | 128K | 0.33 | 0.20 | 0.25 | 0.00 | 0.00 | 0.00 | 0.10 | 1.00 | 0.18 | 0.20 | 0.25 | 0.22 | 0.31 | 0.20 | 0.24 |
| gpt-4-turbo-2024-04-09 | 0.0 | 128K | 0.12 | 0.20 | 0.15 | 0.00 | 0.00 | 0.00 | 1.00 | 0.25 | 0.40 | 0.40 | 0.25 | 0.31 | 0.08 | 0.45 | 0.14 |

Despite having the smallest window of 8K, `llama3-70b` achieves high R across the majority of suites, but suffers lower comparative P scores. The balance between P and R would be an important consideration for practical forensic applications, to minimise the number of false findings that would need to be manually validated by DF practitioners. Finally, our findings suggest that performance could also be highly domain-specific, meaning that different LLMs would perform better in certain analysis contexts.

### B. Limitations

Firstly, we consider our findings preliminary because the analysed data sets represent only a subset of prompt injection attacks and were obtained using a single analysis iteration for each model. Secondly, the test suites used are synthetic, and may not fully represent a set of contextual real-life tasks and workflows. Thirdly, we have not explored the impact of smaller context window sizes and different temperature values, as well as latency and cost considerations, which could impact practical adoption potential. Finally, our approach does not consider the possible impact of anti-forensics, which is especially relevant in the context of IPI attacks. Despite the focus on IPI attacks, we have not explored the applicability of this approach to other types of prompt injection and broader attacks on LLM agents.

## V. CONCLUSION

This paper introduced a DF analysis approach based on LLM agent execution logs, specifically focussed on using LLMs to identify IPI attacks encapsulated in multi-step agentic workflows. Using our experimental approach and a selection of state-of-the-art models, we showcased the promising potential of this method when coupled with LLMs such as `claude-3-opus`, `gpt-4o`, and `gemini-1.5-flash`. Our future work will focus on expanding the validation scope of our approach to address the described limitations, as well as exploring the practical utility of this approach with DF practitioners.

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